Lesson 6: Nothing Is Lost

Unit

Energy: Stores and Transfers

Lesson focus

Students use the principle of conservation of energy to link the three stores they can calculate. They follow energy from one store to another in a closed system, draw energy bar charts, and test the idea with a pendulum.

Specification

AQA GCSE Physics 4.1.2.1 Energy transfers in a system, and 4.1.1.1 (calculating how the total energy is redistributed when a system changes).

Scope

Students calculate the energy in one store and use conservation to state the energy in another. They may find a height from that energy ($h = E_p \div (m\ g)$). They are **not** asked to find a speed from E_k.

Learning objectives

By the end of the lesson, students should be able to:

1. State the principle of conservation of energy.
2. Describe the energy transfers in a closed system, such as a falling ball or a swinging pendulum.
3. Use conservation of energy to calculate the energy in one store from the energy in another.
4. Draw energy bar charts that show the total staying the same.
5. Explain why real systems never quite return to where they started.

Key vocabulary

Keyword	Student-friendly definition
Conservation of energy	Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed
Closed system	A system that no energy enters or leaves; its total energy stays the same
Dissipated	Spread out into the surroundings, usually as thermal energy, where it is less useful
Energy bar chart	A chart showing how much energy is in each store at one moment, drawn to the same scale each time

The principle

> *Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed.*

In a closed system, the total energy before a change equals the total energy after it.

Equipment and resources

- A pendulum (a heavy bob on a string, clamped high)
- Per group: clamp stand, string and a bob of known mass, metre rule, a phone to film the swing; optional ramp, trolley and light gate
- Calculators

Starter: The pendulum that will not hit you

Demonstrate (or show a video): a heavy pendulum bob is held against the teacher's nose and released. It swings away and comes back, stopping just short.

Safety: release the bob from rest and never push it; stand completely still until it has swung back. Rehearse before the lesson, and hang it from a ceiling hook or a firmly clamped stand that cannot tip.

Ask: why does it not come back higher and hit the teacher? Why does it not quite come back as high? Collect answers; the first is conservation, the second is dissipation.

Main teaching: Falling objects

Ignore air resistance for now; the ball and the Earth form a closed system.

Worked example 1:

> *A 0.50 kg ball is dropped from a height of 3.0 m. Calculate the energy in its kinetic store just before it hits the ground. (g = 9.8 N/kg; ignore air resistance)*
> *Energy lost from the gravitational potential store = energy gained by the kinetic store*
> *$E_p = m\ g\ h$*
> *$E_p = 0.50 \times 9.8 \times 3.0$*
> *$E_p = 14.7$ J*
> *So $E_k = 14.7$ J*

Worked example 2 (finding a height):

> *A 0.25 kg ball is thrown straight up with 20 J in its kinetic store. Calculate the greatest height it reaches. Ignore air resistance.*
> *At the top, all 20 J is in the gravitational potential store.*
> *$E_p = m\ g\ h$, so $h = E_p \div (m\ g)$*
> *$h = 20 \div (0.25 \times 9.8)$*
> *$h = 20 \div 2.45$*
> *$h = 8.2$ m*

Worked example 3 (three stores):

> *A toy launcher has a spring with k = 200 N/m, compressed by 0.050 m. It fires a 10 g ball straight up. Calculate the greatest height the ball could reach.*
> *$E_e = ½\ k\ e^2$*
> *$E_e = 0.5 \times 200 \times 0.050^2 = 0.25$ J*
> *At the top, $E_p = 0.25$ J*
> *$h = E_p \div (m\ g)$*
> *$m = 10$ g $= 0.010$ kg*
> *$h = 0.25 \div (0.010 \times 9.8)$*
> *$h = 2.6$ m*

Main teaching: Energy bar charts

Draw a bar for each store at three moments, always to the same scale. For the dropped ball in example 1:

Moment	Gravitational potential	Kinetic	Total
Released	14.7 J	0 J	14.7 J
Half-way down	7.35 J	7.35 J	14.7 J
Just before impact	0 J	14.7 J	14.7 J

Then repeat for a swinging pendulum (top of swing, bottom, other side). Finally, a real pendulum after many swings: a growing thermal bar, the other two shrinking.

Practical: Does the pendulum keep its energy?

Groups test whether a pendulum keeps all its energy. Every reading is a height, so no fast timing is needed.

1. Hang a bob of known mass (for example 0.20 kg) from a clamp, with a metre rule clamped vertically just behind it.
2. Note the height of the bottom of the bob at its lowest point.
3. Pull the bob aside until it is 0.20 m higher than its lowest point. Release it from rest.
4. Read the height it reaches on the far side of its first swing. Filming the swing against the rule with a phone makes this much easier.
5. Let it swing on. Read the height it reaches after 10 swings.
6. Calculate $E_p = m\,g\,h$ at the release height, after one swing and after 10 swings.

Sample results (m = 0.20 kg):

Moment	Height above lowest point / m	E_p / J
Release	0.20	0.39
After 1 swing	0.19	0.37
After 10 swings	0.12	0.24

Discussion:

1. At the lowest point, how much energy is in the kinetic store on the first swing? (About 0.39 J: the energy lost from the gravitational potential store.)
2. Where did the missing energy (about 0.15 J) go after 10 swings? (The thermal store of the air and the string's support, through air resistance and friction.)
3. Is this system closed? Students should conclude that energy was conserved, but some was dissipated each swing. This sets up Lesson 8.

If you have light gates, the ramp version also works: a trolley released from a measured height, its speed measured by a light gate at the bottom, and E_k compared with E_p. Do not time a short run at the bottom with a stopwatch: reaction time is a large fraction of the time, and the error in E_k is larger still, because the speed is squared.

Check for understanding

1. State the principle of conservation of energy.
2. A 2.0 kg rock falls 5.0 m. How much energy is in its kinetic store when it lands? Ignore air resistance. (98 J)
3. A 0.10 kg ball is thrown up with 4.9 J in its kinetic store. How high does it go? (5.0 m)
4. In the practical, the pendulum swung lower each time. Does this break the principle of conservation? Explain.

Plenary

Return to the pendulum. Students write two sentences: one explaining why it does not hit the teacher, one explaining why each swing is lower.

Possible prep

Draw energy bar charts for a rollercoaster car at the top of the first hill, at the bottom, and at the top of a smaller second hill.

Teacher notes

1. Insist on "energy is transferred from … to …". Students often write "gravitational energy turns into kinetic energy". Accept it early on, then upgrade the language.
2. Air resistance is a force, not a store. The energy it transfers goes to the thermal store of the air.
3. Students may notice that the bob's or ball's mass cancels when finding h. Do not develop this; it is a good stretch question for students who spot it.
4. The pendulum loses only a little on each swing, so the one-swing reading may look like no loss at all. The 10-swing reading shows the trend clearly. Do not present the shortfall as a failed experiment: it is the evidence for dissipation.